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# Ganoderma Lucidum Alleviates Ischemia-Reperfusion–Induced Renal Injury in Wistar Rats

Ahmet Dönder, Ph.D., and Fırat Aşır, Ph.D.

**OBJECTIVE:** To investigate effects of *Ganoderma lucidum* (*G. lucidum*) kidney sections of rats induced by ischemia-reperfusion (I/R) injury.

**STUDY DESIGN:** A total of 40 rats were assigned to 4 groups: control (Sham), I/R, *G. lucidum*, and I/R+*G. lucidum* groups. Prior to animal experiments, 20 mL/kg *G. lucidum* was administered to the *G. lucidum*–treated groups for 7 days. The control and I/R groups received only saline solution. The kidney was exposed to hypoxia for 1 hour by clamping renal vessels and was then allowed to reoxygenate for 6 hours. Blood was taken to measure for serum MDA, MPO, and GSH. Kidney tissues were resected for histological paraffin tissue protocol. Hematoxylin-eosin and immunohistochemical staining were performed.

**RESULTS:** MDA and MPO levels were highest in the I/R group but were close to the levels of the control group in the I/R+*G. lucidum* group. Unlike MDA and MPO values, GSH values were the lowest in the I/R group, but after *G. lucidum* treatment, GSH levels increased in the I/R+*G. lucidum* group. In kidney sections of hematoxylin-eosin staining, the control group showed no pathology. In the I/R group, atrophic glomeruli, degenerated tubular cells, and mononuclear cell infiltration with dilated and congested vessels were observed. In the I/R+*G. lucidum* group, I/R pathol-

ogy was mostly recovered. In the *G. lucidum* group, ADAMTS-4 expression was moderately expressed in glomerular and tubular cells. The I/R group showed positive ADAMTS-4 expression in mostly inflammatory cells. In the I/R+*G. lucidum* group, ADAMTS-4 was positively expressed only in glomerular structures. In the *G. lucidum* group, caspase-3 expression was observed in glomerular and tubular cells. The I/R group showed strong caspase-3 activity in glomerular and tubular cells, in vascular cells, and inflammatory cells. The I/R+*G. lucidum* group showed weak caspase-3 expression.

**CONCLUSION:** Ischemia-reperfusion injury caused histopathological and biochemical alterations in renal tissue; *G. lucidum* protected tissue structure and integrity by its antioxidant and anti-inflammatory properties. (Anal Quant Cytopathol Histopathol 2021;43: 484–492)

**Keywords:** Chinese herbal drugs, *Ganoderma lucidum*, ischemia-reperfusion injury, Chinese traditional medicine, renal ischemia-reperfusion.

Renal ischemia-reperfusion (I/R) injury occurs when blood flow to the kidney is restricted and then allowed to reoxygenate. Infarction, shock,

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**Financial Disclosure:** The authors have no connection to any companies or products mentioned in this article.

sepsis, and kidney transplantation are common causes of renal I/R injury, leading to tissue damage.<sup>1,2</sup> During ischemia, continuous oxygen supplied to tissues is blocked from maintaining normal physiological function; therefore, restorative blood flow (reperfusion) is important to compensate for ischemic cell death.<sup>3,4</sup> Ischemia-reperfusion injury involves complex pathological processes such as an inflammatory cascade including reactive oxygen species (ROS),<sup>5</sup> cytokines, chemokines, and leukocyte activation.<sup>6,7</sup> Renal I/R disrupts calcium homeostasis, prevents phospholipase and protease activation, and changes cellular pH.<sup>8,9</sup> These alterations lead to tissue damage, eventually causing renal damage and renal dysfunction.<sup>8</sup> When the ischemia period is extended, acute tubular necrosis and glomerular damage may occur and give rise to the dysfunction of grafts and rejection in the transplantation process. In cases of severe ischemia period, chronic kidney disease, end-stage renal disease, and even death can occur.<sup>10</sup>

In order to overcome clinical complications of renal I/R, several agents were alternatively introduced in the studies. Montelukast,<sup>11</sup> resveratrol,<sup>12</sup> Ginkgo biloba extract,<sup>13</sup> and luteolin<sup>14</sup> are just a few of the agents used to treat renal I/R. Ganoderma lucidum (G. lucidum) is a white-rot fungus belonging to the Ganoderma genus, Ganodermataceae family, that has been used in traditional medicine for years, especially in Asian countries.<sup>15</sup> G. lucidum is used in many studies due to its curative properties in many diseases such as cancer, immunologic disorders, and neurodegenerative diseases.<sup>16,17</sup> Fruiting bodies of G. lucidum include many chemical compounds such as polysaccharides, triterpenoids, and proteins. In one study G. lucidum peptides protected rat liver tissue and mitochondrial membrane peroxidation systems with its antioxidant, metal-chelating, and free radical-scavenging activities.<sup>18</sup>

In this study we aimed to investigate the protective effect of G. lucidum by its antioxidant activities on Wistar albino rat kidney tissue against renal I/R injuries.

## Materials and Methods

### Experimental Design and G. Lucidum Preparation

Ethical approval was obtained from Dicle University Health Sciences Application and Research Center Ethics Committee. Animal experimentation was conducted in accordance with Dicle University Animal Care and Use Committee. The G. lucidum

fungus mixture (water soluble) was imported from Shandong Si Wei Co., Ltd. (Heze, Shandong Province, China) (license no. Z200220083). A pure culture of G. lucidum mycelia was inoculated into medium containing bagasse and defatted rice bran and allowed to culture for 3–4 months until fruit body was developed. Then, fruit body was air-dried, extracted with boiling water, and filtrated by the method described by Zhang et al.<sup>19</sup> Wistar albino rats (220–250 g, male) were housed in a well-aerated individual cage in an animal room with 12-hour light/dark cycles,  $24 \pm 2^\circ\text{C}$ , and 10–55% humidity. The rats had free access to water and commercial chow *ad libitum*.

The animals were divided into 4 groups, 10 animals per group. All procedures on the rats were performed under anesthesia. G. lucidum was prepared every day and freshly administered to the rats as follows.

**Control (Sham) Group.** Prior to the experiment, 20 mL/kg 0.9% saline solution was orally administered to the rats for 7 days. The abdomen was shaved and sterilized with 10% polyvinylpyrrolidone iodine complex (Adeka, Istanbul, Turkey). Approximately 3 cm of skin incision was opened on the abdominal midline. Then the incision was sutured.

**G. Lucidum Group.** Prior to the experiment, 20 mL/kg G. lucidum was orally administered to the rats for 7 days. The abdomen was opened and closed.

**Ischemia-Reperfusion (I/R) Group.** Prior to the experiment, 20 mL/kg 0.9% saline solution was orally administered to the rats for 7 days. After the abdomen was opened, the renal artery and vein were clamped for 1 hour to create ischemia. Then, blood was reperfused for 6 hours and the abdomen was sutured.

**Ischemia-Reperfusion+G. Lucidum (I/R+G. Lucidum) Group.** Prior to the experiment, 20 mL/kg G. lucidum was administered to the rats for 7 days by orogastric route. After the abdomen was opened, the renal artery and vein were clamped for 1 hour to create ischemia. Then, blood was reperfused for 6 hours and the abdomen was sutured. At the end of the experiment, all animals were sacrificed under anesthesia. Blood samples were collected for oxidative stress analysis. The kidney tissues were resected for histopathological analysis.

### Biochemical Analyses

Blood samples were obtained from the heart of animals to measure serum malondialdehyde (MDA), glutathione (GSH), and myeloperoxidase (MPO) levels. The blood samples were centrifuged for 5 minutes at 1,550 g and supernatant was discarded. The MDA level was measured with the method described by Cavus et al.<sup>20,21</sup>

### Histological Tissue Processing

Histological tissue processing was applied according to Dag et al and Gem et al.<sup>22,23</sup> Kidney tissues were fixed with zinc-formalin solution (catalog no. Z2902, Sigma-Aldrich, St. Louis, Missouri, USA). Tissues were passed through ascending alcohol series and washed in xylene and incubated within paraffin wax. 5 µm sections were cut, deparaffinized in xylene, and brought to distilled water. Some of the sections were stained with routine hematoxylin and eosin, and the rest were kept for immunohistochemical staining.

### ADAMTS-4 and Caspase-3 Immunostaining

Immunostaining was done by the method of Ozkorkmaz et al and Ozevren et al.<sup>24,25</sup> Hydrogen peroxide solution (catalog no. TA-015-HP, Thermo Fisher, Fremont, California, USA) was dropped on sections for 20 minutes. For blocking, Ultra V Block (catalog no. TA-015-UB, Thermo Fisher) was applied to sections for 8 minutes. Sections were incubated with primary antibodies anti-ADAMTS-4 (catalog no. ab219548, Abcam, Cambridge, USA) and anti-caspase-3 (catalog no. ab4051, Abcam) at 4°C overnight. Sections were washed with biotinylated secondary antibody (catalog no. TP-015-BN, Thermo Fisher) for 14 minutes. Streptavidin-peroxidase (catalog no. TS-015-HR, Thermo Fisher) was dropped onto sections for 15 minutes. DAB (catalog no. TA-001-HCX, Thermo Fisher) was used as chromogen. Sections were counterstained with

Harris hematoxylin and mounted with Entellan (catalog no. 107961, Sigma-Aldrich). Slides were analyzed with Zeiss Imager A2 Zen 3.0 software (Germany) and photomicrographed.

### Statistical Analysis

All the data were recorded as mean ± standard deviation. All statistical analyses were calculated with IBM SPSS software version 25. Within-groups comparisons were compared Kruskal-Wallis test and Dunn-Bonferroni post-hoc. Significance level was accepted as  $p < 0.05$ .

## Results

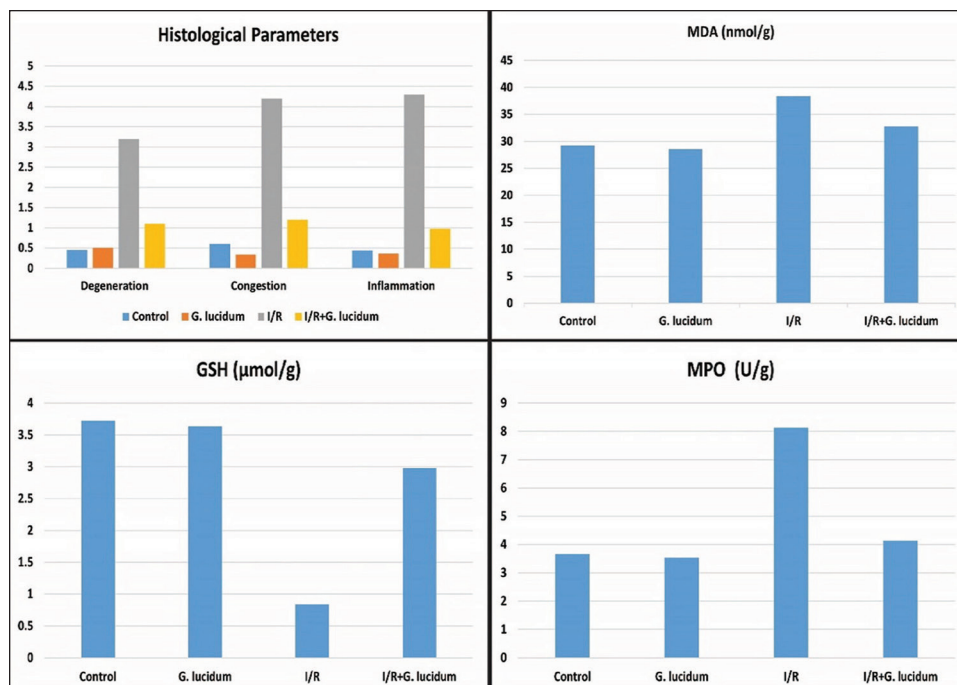
### Biochemical Results

The MDA, GSH, and MPO values were compared within groups (Table I). There were no significant differences between the control and G. lucidum-treated groups regarding all of these biochemical parameters. However, MDA, which is a lipid peroxidation index, was higher in the I/R group than in the control and G. lucidum treated groups. The I/R+G. lucidum group had lower MDA levels than did the I/R group, and the values were close to that of the control group. GSH levels were the lowest in the I/R within the groups. On the contrary, after G. lucidum treatment, GSH levels increased in the I/R+G. lucidum group. Similar to GSH values, the ischemia-reperfusion group had the lowest MPO values. In the I/R+G. lucidum group, the MPO levels were close to those of the control group, showing G. lucidum provided protection by provoking neutrophil infiltration. I/R injury caused elevation of MDA and MPO values and reduction in GSH values due to oxidative damage. These parameters in the I/R+G. lucidum group were close to those of the control group, indicating that G. lucidum treatment protected the liver against oxidative damage. Figure 1 shows graphical illustration of Table I.

**Table I** Histological and Biochemical Parameters by Group

| Group          | Histopathological parameter |            |              | Biochemical parameter |              |           |
|----------------|-----------------------------|------------|--------------|-----------------------|--------------|-----------|
|                | Degeneration                | Congestion | Inflammation | MDA (nmol/g)          | GSH (µmol/g) | MPO (U/g) |
| Control        | 0.45                        | 0.60       | 0.44         | 29.3                  | 3.72         | 3.66      |
| G. lucidum     | 0.50                        | 0.34       | 0.36         | 28.6                  | 3.64         | 3.54      |
| I/R            | 3.20                        | 4.20       | 4.30         | 38.4                  | 0.84         | 8.12      |
| I/R+G. lucidum | 1.10                        | 1.20       | 0.98         | 32.8                  | 2.98         | 4.14      |

I/R = ischemia-reperfusion.



**Figure 1**  
Graphical illustration of Table I regarding histological (degeneration, congestion, and inflammation) and biochemical (MDA, GSH, and MPO values) parameters by group.

### Histopathological Results

In the kidney sections of hematoxylin-eosin staining, the control group showed regular kidney histology. Glomeruli were structurally normal, and nuclei of distal and proximal tubule cells appeared chromatin-rich. No pathology was observed (Figure 2A). In the G. lucidum group, there was mononuclear cell infiltration in the cortex and fibrosis between tubules (Figure 2B). The I/R group showed atrophic glomeruli, degenerated tubular cells, and mononuclear cell infiltration with dilated and congested vessels (Figure 2C). In the I/R+G. lucidum group, severe pathology was not observed except for lessened mononuclear cell infiltration (Figure 2D).

### Immunostaining Results

In ADAMTS-4 immunostaining of kidney sections, the control group showed positive ADAMTS-4 expression mostly in glomerular cells, while negative expression was mostly in tubular cells (Figure 3A). In the G. lucidum group, parietal membrane of glomeruli and tubular cells moderately expressed ADAMTS-4; however, visceral membrane of glomeruli had negative expression (Figure 3B). The ischemia-reperfusion group showed positive ADAMTS-4 expression in glomeruli and its

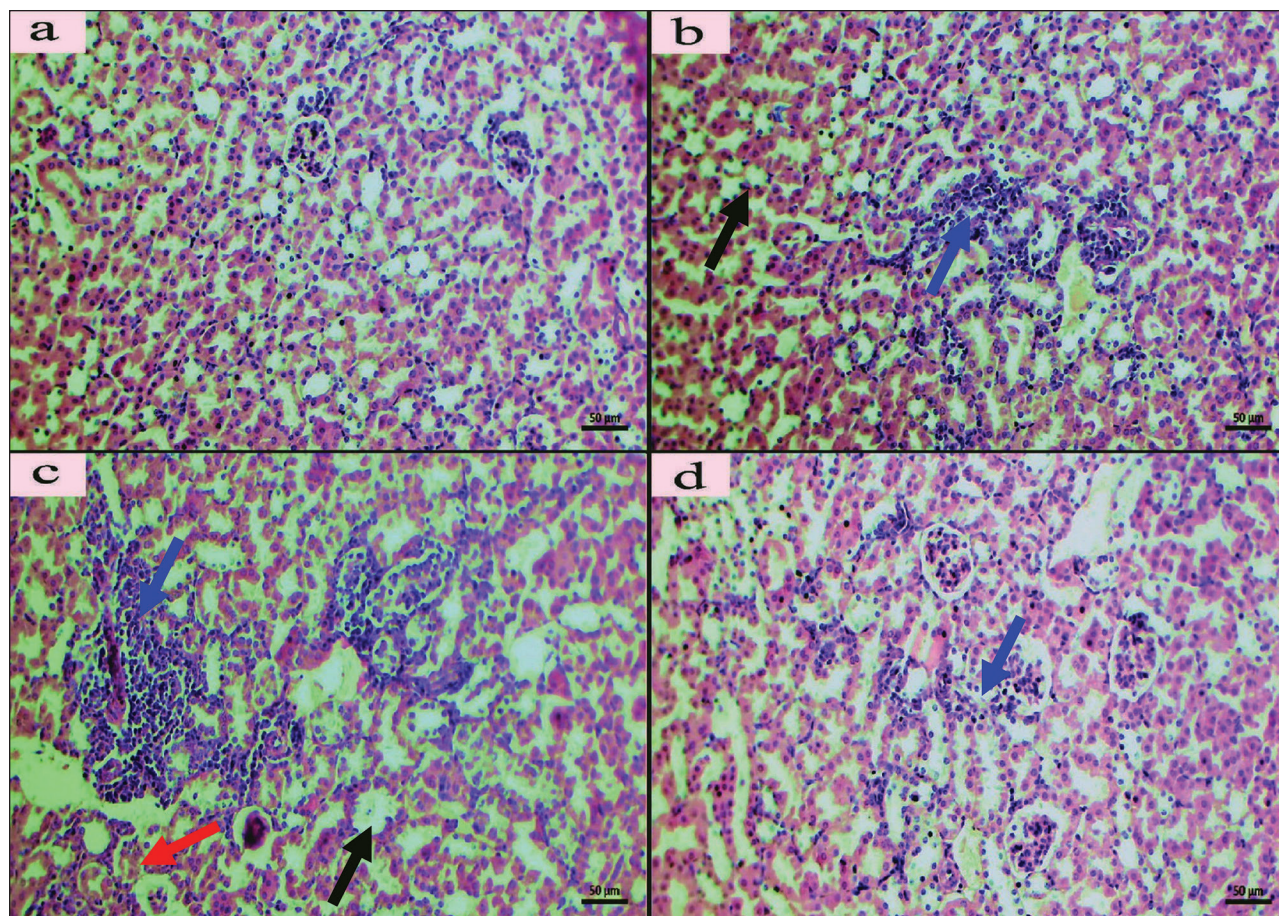
related structures and in inflammatory cells (Figure 3C). In the I/R+G. lucidum group, ADAMTS-4 expression was positively observed in glomerular structures but negatively in some proximal and distal tubules (Figure 3D).

In caspase-3 immunostaining of kidney sections, the control group showed weak caspase-3 expression in glomerular and tubular cells but strong expression in some tubular cells (Figure 4A). In the G. lucidum group, glomerular and tubular cells expressed caspase-3 expression (Figure 3B). The I/R group showed positive caspase-3 expression in glomerular and tubular cells, in vascular cells, and in inflammatory cells (Figure 4C). In the I/R+G. lucidum group, caspase-3 expression was weakly observed in glomerular and tubular cells (Figure 4D).

### Discussion

Many therapeutic agents have been used to show its renal protective effects in studies. Yang et al showed that dietary nitrate protected renal tissues against renal I/R injury by altering oxidative stress and inflammatory response mechanisms, thus suggesting nitrate supplementation as a potential treatment in patients at high risk of renal I/R injury.<sup>26</sup> A study by Sener et al<sup>11</sup> sug-



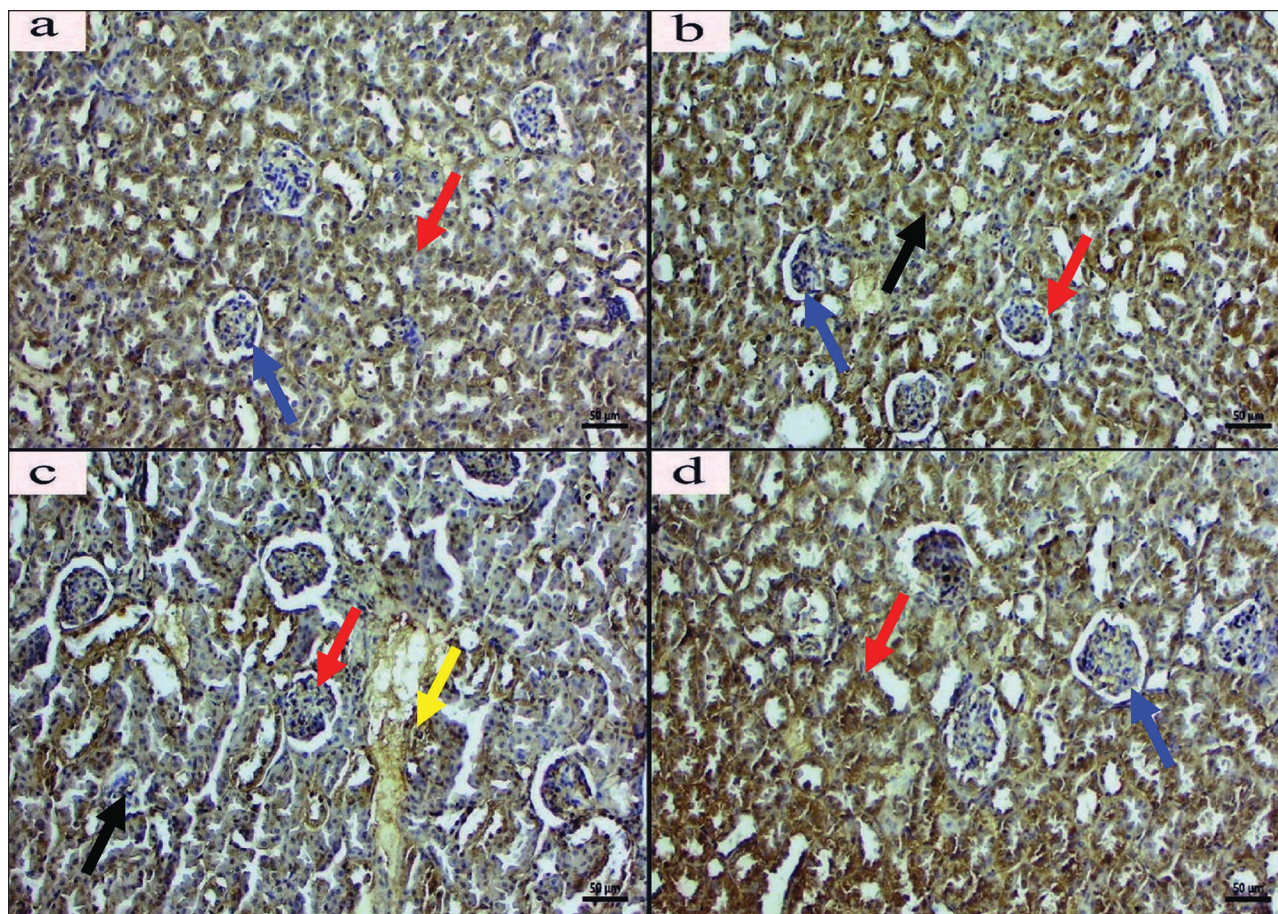


**Figure 2** Hematoxylin-eosin staining of kidney sections. (A) Control group: regular glomeruli in the Bowman capsule, the nuclei of the proximal tubules and distal tubules rich in chromatin, and regular basement membrane borders were observed. (B) *G. lucidum* group: mononuclear cell infiltration around some glomeruli (blue arrow), hypertrophy in some proximal tubular cells (black arrow), and increased intertubular connective tissue were observed. (C) I/R group: shrinkage of renal glomeruli, degeneration in proximal and distal tubular cells (black arrow), inflammatory cell infiltration between tubules (blue arrow), and dilation/congestion in blood vessels (red arrow) were observed. (D) I/R + *G. lucidum* group: regular glomerular structures, distal/proximal tubules with increased chromatin density, and a few inflammatory cells between tubules (blue arrow) were observed.

gested that montelukast, a selective antagonist of cysteinyl leukotriene receptor, improved renal histopathology and altered I/R-induced oxidant responses, inferring that montelukast has renoprotective effects by inhibiting neutrophil infiltration, balancing oxidant-antioxidant status, and regulating the generation of inflammatory mediators. Another study by the same researchers also revealed that resveratrol, the natural phytoalexin present in grapes, peanuts, mulberries, and red wine, protects renal tissue by its radical scavenging and antioxidant activities with prevention of tissue neutrophil infiltration.<sup>12</sup> A study on reactive

oxygen species and renal I/R found that *Ginkgo biloba* extract protects renal tissue and plays a causal role in I/R-induced renal injury and by the radical scavenging and antioxidant activities. They showed that the extract-treated group had lower MDA and MPO values and higher GSH value.<sup>13</sup> Tuğtepe et al<sup>27</sup> studied the effects of oxytocin on renal I/R by histological examination and biochemical parameters. They showed that renal I/R injury elevated MDA, MPO, and reactive oxygen radical levels in the renal tissue, while decreasing renal GSH content. They stated that administration of oxytocin improved renal function and





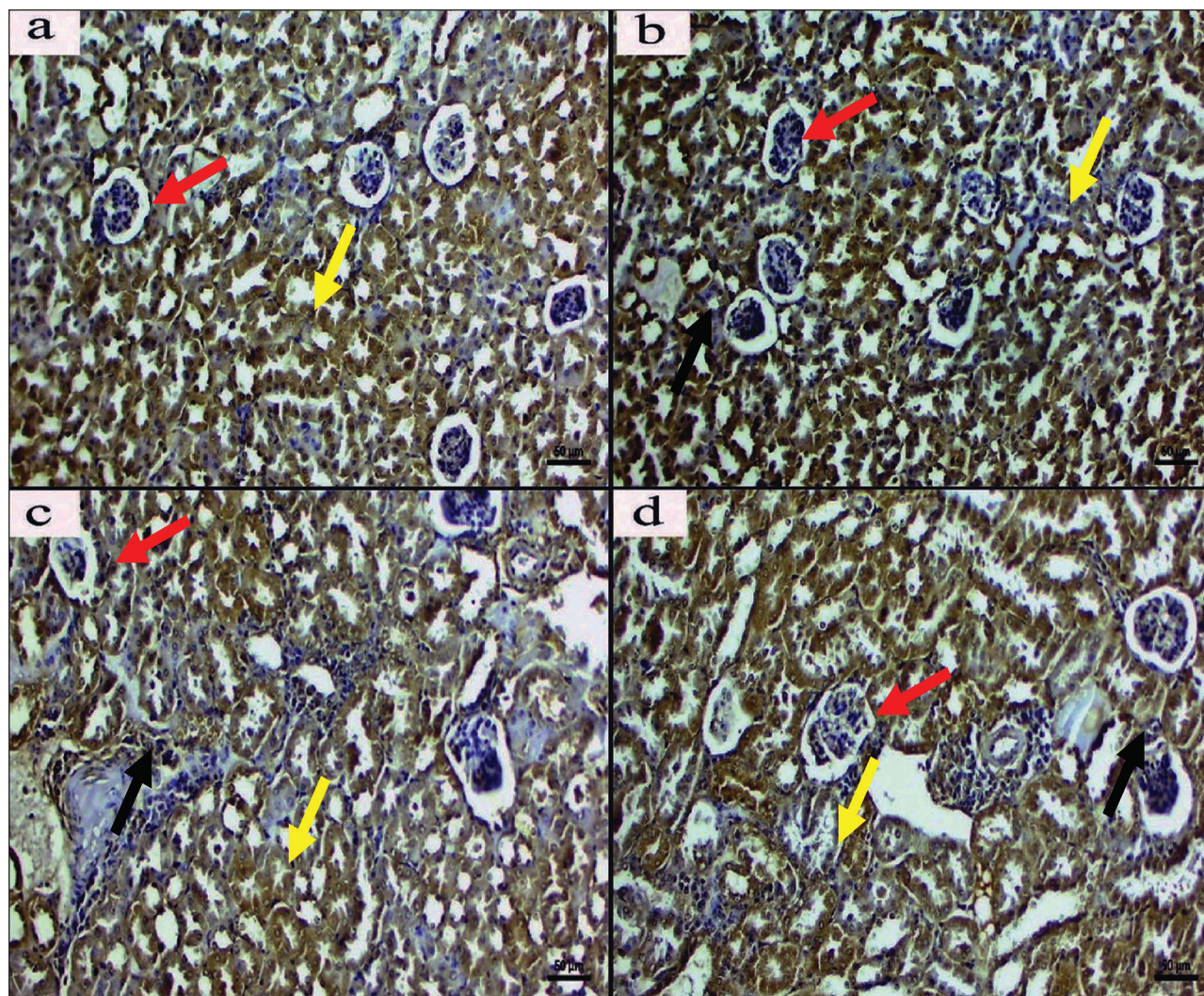
**Figure 3** ADAMTS-4 immunostaining of kidney sections. (A) Control group: ADAMTS-4 expression was negative in cells in the glomerular parietal and visceral membranes (blue arrow), while it was positive in some proximal and distal tubular cells (red arrow). (B) G. lucidum group: parietal membranes of the glomeruli (red arrow) and proximal/distal tubules (black arrow) showed moderate ADAMTS-4 expression; however, negative expression was observed in the visceral membranes (blue arrow). (C) I/R group: ADAMTS-4 was positively expressed in inflammatory cells around dilated and conjugated vessels (yellow arrow), visceral and parietal membranes of glomeruli (red arrow), macula densa, and most proximal tubules (black arrow). (D) I/R + G. lucidum group: while negative ADAMTS-4 expression was observed in the visceral and parietal membranes of the glomeruli (blue arrow), it was positive in the nuclei of most proximal and distal cells (red arrow).

microscopic damage, along with the alleviation of oxidant tissue responses, suggesting that oxytocin protects renal tissue against I/R-induced oxidative damage. Hong et al<sup>14</sup> researched luteolin's anti-inflammatory and antioxidant effects and showed that it protects renal tissue against renal I/R in rats by reducing oxidative stress, neutrophil infiltration, inflammation, renal cell apoptosis, and expression of HMGB1 and NF- $\kappa$ B, and ER stress. In our study, we used G. lucidum polysaccharide peptide as an agent against renal I/R injury. Our biochemical results showed that I/R injury caused an elevation in MDA and MPO val-

ue and a decrease in GSH content. However, the MDA and MPO values decreased and GSH content increased in the I/R group treated with G. lucidum. Our biochemical results suggest that G. lucidum alters tissue oxidative response by its radical scavenging and its antioxidant activities.

Renal I/R injury results from blockage of oxygen and nutrient delivery to kidney tissues and removal of waste products from kidney cells.<sup>28</sup> Once this balance is disrupted, damage to tubular epithelial cells occurs, leading to death of the cells by apoptosis and necrosis. Then, water and electrolyte homeostasis of the kidney is impaired





**Figure 4** Caspase-3 immunostaining of kidney sections. (A) Control group: caspase-3 expression was weakly observed in the glomerular and tubular cells (red arrow), and strongly in a small number of cells in some small tubules (yellow arrow). (B) *G. lucidum* group: poor caspase-3 expression was seen in glomerular (red arrow) and tubular (yellow arrow) cells and in macula densa (black arrow). (C) I/R group: caspase-3 expression was positive in glomerular and degenerative cells around the glomeruli (red arrow), most proximal/distal tubule cells (yellow arrow), and dilated and congested vascular endothelial cells and surrounding inflammatory cells (black arrow). (D) I/R+*G. lucidum* group: caspase-3 expression was weakly observed in glomerular (red arrow) and tubular (yellow arrow) cells and in macula densa (black arrow).

and excretion of waste products is decreased. The physiopathology of this process is complex and involves hemodynamic changes, inflammation, and endothelial and epithelial cell injury, followed by repair that can be adaptive and restore epithelial integrity or maladaptive, leading to chronic kidney disease.<sup>7</sup> In their study, Zhong et al<sup>29</sup> demonstrated that renal I/R caused tubular brush border loss and vascular dilation and intertubular hem-

orrhage. They found that *G. lucidum* ameliorated these effects of renal I/R by reducing oxidative stress and alleviating the mitochondrial and ER stress-dependent apoptosis caused by excessive ROS. In our I/R group, atrophied glomeruli, degenerated proximal and distal tubular cells, inflammatory cell infiltration between tubules, and dilation and congestion and apoptotic changes in blood vessels were observed (Figure 2C). In the



I/R+G. lucidum group, due to strong antioxidant activity of G. lucidum, there were regular glomeruli structures. The chromatin-rich nuclei of tubular cells were increased, and there were also a few solitary inflammatory cells among the tubular structures (Figure 2D).

ADAMTS-4 is a member of an enzyme belonging to ADAMTS (A disintegrin and metalloproteinase with thrombospondin motifs) protein family. It plays a role in degradation of some proteoglycans including aggrecan, neurocan, and versican.<sup>30,31</sup> ADAMTS-4 expression is observed in many organs such as ovary, spinal cord, adrenal cortex, brain, retina, and pancreas (islets).<sup>32</sup> ADAMTS-4 expression is mainly related to articular cartilage in osteoarthritis.<sup>33</sup> Lemarchant et al<sup>34</sup> found that ADAMTS-4 reduced neuroinflammation in vitro and in vivo after ischemic stroke, suggesting that ADAMTS-4 may be used to treat acute central nervous system injuries such as cerebral ischemia, ischemic stroke, or traumatic brain injury due to its anti-inflammatory and neuroregenerative properties. Another study revealed that ADAMTS-4 expression is upregulated in astrocytes by induction of TNF in vivo following cerebral artery stroke.<sup>5</sup> A study conducted by Wagsater et al found that inflammatory enzyme ADAMTS-4 with ADAMTS-8 expression increases in macrophage-rich areas of atherosclerotic plaques.<sup>35</sup> In our study ADAMTS-4 expression was mainly observed in inflammatory cells and also in vessels and glomerular structures in the I/R group. In the G. lucidum-treated group with I/R, the expression was observed in glomerular membranes and tubular cells. These findings state that ADAMTS-4 expression is upregulated due to inflammation mechanism after I/R. G. lucidum showed renoprotective effects after I/R due to its high antioxidant activity.

Caspase-3 is a member of the caspase family that interacts with caspase-8 and caspase-9, playing a role in apoptosis. Activated caspase-3 can be involved in many vital cellular events by cleaving key regulatory proteins for the cells.<sup>36</sup> In the study of Yang et al, they found that caspase-3 is an important regulator of microvascular endothelial cell apoptosis and renal fibrosis after I/R since long-term silencing of caspase-3 gene causes fibrosis and reduced microvascular rarefaction.<sup>37</sup> In a study of cerebral I/R, caspase-3 expression was markedly increased within 24 hours of reperfusion after cerebral ischemia. The authors suggest that caspase-3 activation contributes to cell death

after ischemia with reperfusion.<sup>38</sup> Liu et al stated that myocardial I/R injury increased caspase-3 expression and apoptosis; however, injection of traditional Chinese medicine (Shengmai) decreased ischemia-induced cardiomyocyte apoptosis, with decreased expression of caspase-3.<sup>39</sup> In our caspase-3 immunostaining of kidney sections, the I/R group showed strong caspase-3 expression in glomerular structures, degenerative cells around the glomeruli, tubular cells, vascular endothelial cells, and surrounding inflammatory cells. However, caspase-3 expression was weakly observed in the I/R+G. lucidum group, suggesting that G. lucidum blocks the apoptosis mechanism by indirectly inhibiting the pathway of caspase-3, therefore promoting cell survival and preventing cell death.

In conclusion G. lucidum showed a renoprotective effect against I/R injury on the rat kidney by its antioxidant and anti-inflammatory properties.

## References

1. Jang HR, Rabb H: The innate immune response in ischemic acute kidney injury. *Clin Immunol* 2009;130(1):41-50
2. Sharfuddin AA, Molitoris BA: Pathophysiology of ischemic acute kidney injury. *Nat Rev Nephrol* 2011;7(4):189-200
3. Bellomo R, Kellum JA, Ronco C: Acute kidney injury. *Lancet* 2012;380(9843):756-766
4. Thadhani R, Pascual M, Bonventre JV: Acute renal failure. *N Engl J Med* 1996;334(22):1448-1460
5. Cross AK, Haddock G, Stock CJ, Allan S, Surr J, Bunning RA, Buttle DJ, Woodroffe MN: ADAMTS-1 and -4 are up-regulated following transient middle cerebral artery occlusion in the rat and their expression is modulated by TNF in cultured astrocytes. *Brain Res* 2006;1088(1):19-30
6. Bonventre JV, Zuk A: Ischemic acute renal failure: An inflammatory disease? *Kidney Int* 2004;66(2):480-485
7. Bonventre JV, Yang L: Cellular pathophysiology of ischemic acute kidney injury. *J Clin Invest* 2011;121(11):4210-4221
8. Paller MS: The cell biology of reperfusion injury in the kidney. *J Invest Med* 1994;42(4):632-639
9. Weight SC, Bell PR, Nicholson ML: Renal ischaemia-reperfusion injury. *Br J Surg* 1996;83(2):162-170
10. Liano F, Pascual J: Epidemiology of acute renal failure: A prospective, multicenter, community-based study. Madrid Acute Renal Failure Study Group. *Kidney Int* 1996;50(3):811-818
11. Sener G, Sehirli O, Velioglu-Ogünç A, Cetinel S, Gedik N, Caner M, Sakarcan A, Yeğen BC: Montelukast protects against renal ischemia/reperfusion injury in rats. *Pharmacol Res* 2006;54(1):65-71
12. Sener G, Tuğtepe H, Yüksel M, Cetinel S, Gedik N, Yeğen BC: Resveratrol improves ischemia/reperfusion-induced oxidative renal injury in rats. *Arch Med Res* 2006;37(7):822-829
13. Sener G, Sener E, Sehirli O, Ogünç AV, Cetinel S, Gedik

- N, Sakarcı A: Ginkgo biloba extract ameliorates ischemia reperfusion-induced renal injury in rats. *Pharmacol Res* 2005; 52(3):216-222
14. Hong X, Zhao X, Wang G, Zhang Z, Pei H, Liu Z: Luteolin treatment protects against renal ischemia-reperfusion injury in rats. *Mediators Inflamm* 2017;2017:9783893
  15. Ji Z, Tang Q, Zhang J, Yang Y, Jia W, Pan Y: Immunomodulation of RAW264.7 macrophages by GLIS, a proteopolysaccharide from *Ganoderma lucidum*. *J Ethnopharmacol* 2007; 112(3):445-450
  16. Cheung WM, Hui WS, Chu PW, Chiu SW, Ip NY: *Ganoderma* extract activates MAP kinases and induces the neuronal differentiation of rat pheochromocytoma PC12 cells. *FEBS Lett* 2000;486(3):291-296
  17. Wasser SP, Weis AL: Therapeutic effects of substances occurring in higher Basidiomycetes mushrooms: A modern perspective. *Crit Rev Immunol* 1999;19(1):65-96
  18. Sun J, He H, Xie BJ: Novel antioxidant peptides from fermented mushroom *Ganoderma lucidum*. *J Agric Food Chem* 2004;52(21):6646-6652
  19. Zhang W, Zhang Q, Deng W, Li Y, Xing G, Shi X, Du Y: Neuroprotective effect of pretreatment with *ganoderma lucidum* in cerebral ischemia/reperfusion injury in rat hippocampus. *Neural Regen Res* 2014;9(15):1446-1452
  20. Çavuş Y, Deger U: Investigation of ADAMTS-1, BAX and IL-10 expressions in granulosa cells determining fertilization success in IVF. *Int J Morphol* 2020;38(2):427-434
  21. Deger U, Çavuş Y: Morphometric and immunohistochemical examination of changes in umbilical cord structure of placenta previa and normotensive patients. *Anal Quant Cytopathol Histopathol* 2020;42:49-53
  22. Dag U, Deveci E: Protective effect of losartan, an angiotensin type 1 receptor blocker, on retinal damage caused by traumatic brain injury. *Anal Quant Cytopathol Histopathol* 2021;43(3):121-128
  23. Gem M, Sahin I, Uzel K, Ermis IS, Deveci E: Effect of graft application and nebulol treatment on tibial bone defect in rats. *Anal Quant Cytopathol Histopathol* 2021;43(4):223-228
  24. Gokalp-Ozkorkmaz E, Asir F, Basaran SO, Agacayak E, Sahin F, Kaya S, Erdogan G, Deveci E: Examination of Bcl-2 and Bax protein levels for determining the apoptotic changes in placentas with gestational diabetes and preeclampsia. *Proceedings* 2018;2(25):1548
  25. Özevren H, İrtegin S, Deveci E, Aşır F, Pektanç G, Deveci S: *Ganoderma lucidum* protects rat brain tissue against trauma-induced oxidative stress. *Korean J Neurotrauma* 2017;13(2):76-84
  26. Yang T, Zhang XM, Tarnawski L, Peleli M, Zhuge Z, Ter-rando N, Harris RA, Olofsson PS, Larsson E, Persson AEG, Lundberg JO, Weitzberg E, Carlstrom M: Dietary nitrate attenuates renal ischemia-reperfusion injuries by modulation of immune responses and reduction of oxidative stress. *Redox Biol* 2017;13:320-330
  27. Tuğtepe H, Sener G, Biyikli NK, Yüksel M, Cetinel S, Gedik N, Yeğen BC: The protective effect of oxytocin on renal ischemia/reperfusion injury in rats. *Regul Pept* 2007;140(3):101-108
  28. Chatterjee PK: Novel pharmacological approaches to the treatment of renal ischemia-reperfusion injury: A comprehensive review. *Naunyn Schmiedebergs Arch Pharmacol* 2007;376(1-2):1-43
  29. Zhong D, Wang H, Liu M, Li X, Huang M, Zhou H, Lin S, Lin Z, Yang B: *Ganoderma lucidum* polysaccharide peptide prevents renal ischemia reperfusion injury via counteracting oxidative stress. *Sci Rep* 2015;5:16910
  30. Tang BL, Hong W: ADAMTS: A novel family of proteases with an ADAM protease domain and thrombospondin 1 repeats. *FEBS Lett* 1999;445(2-3):223-225
  31. Oglak SC, Obut M: Expression of ADAMTS13 and PCNA in the placentas of gestational diabetic mothers. *Int J Morphol* 2021;39(1):38-44
  32. Kelwick R, Desanlis I, Wheeler GN, Edwards DR: The ADAMTS (A Disintegrin and Metalloproteinase with Thrombospondin motifs) family. *Genome Biol* 2015;16:113
  33. Bondeson J, Wainwright S, Hughes C, Caterson B: The regulation of the ADAMTS4 and ADAMTS5 aggrecanases in osteoarthritis: A review. *Clin Exp Rheumatol* 2008;26(1):139-145
  34. Lemarchant S, Dunghana H, Pomeschik Y, Leinonen H, Kolosowska N, Korhonen P, Kanninen KM, García-Berrocso T, Montaner J, Malm T, Koistinaho J: Anti-inflammatory effects of ADAMTS-4 in a mouse model of ischemic stroke. *Glia* 2016;98(9):1492-1507
  35. Wågsäter D, Björk H, Zhu C, Björkegren J, Valen G, Hamsten A, Eriksson P: ADAMTS-4 and -8 are inflammatory regulated enzymes expressed in macrophage-rich areas of human atherosclerotic plaques. *Atherosclerosis* 2008;196(2):514-522
  36. Porter AG, Janicke RU: Emerging roles of caspase-3 in apoptosis. *Cell Death Differ* 1999;6(2):99-104
  37. Yang B, Lan S, Dieudé M, Sabo-Vatasescu JP, Karakeussian-Rimbaud A, Turgeon J, Qi S, Gunaratnam L, Patey N, Hébert MJ: Caspase-3 is a pivotal regulator of microvascular rarefaction and renal fibrosis after ischemia-reperfusion injury. *J Am Soc Nephrol* 2018;29(7):1900-1916
  38. Manabat C, Han BH, Wendland M, Derugin N, Fox CK, Choi J, Holtzman DM, Ferriero DM, Vexler ZS: Reperfusion differentially induces caspase-3 activation in ischemic core and penumbra after stroke in immature brain. *Stroke* 2003;34(1):207-213
  39. Liu X, Tan W, Yang F, Wang Y, Yue S, Wang T, Wang X: Shengmai injection reduces apoptosis and enhances angiogenesis after myocardial ischaemia and reperfusion injury in rats. *Biomed Pharmacother* 2018;104:629-636